

Towards VWA of Affective Noise in SET

An interpretable ABSA framework that adjusts SET ratings by down-weighting construct-irrelevant emotion.



17,127



RateMyProfessors Reviews



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- Wilfrid Laurier University, Ontario, Canada
- August 2026
- Special Session: SS-MFRA



RMP

🍏 Professors ▾

Professor name

? / 5

Overall Ratings Based on a Biased System

**RateMy
Professor** 📖

100%

Chance of it staying here for a long
time

1.0

Degree of
fairness

Rate →

Compare

Rating Distribution

Awesome 5



Great 4



Good 3



OK 2



Awful 1



QUALITY

3.0

Oct 30th, 2008

Textbook: **No**

DIFFICULTY

5.0

dude is huge and i mean huge and is a rude guy but he knows alot about politics

Helpful  1  2



QUALITY

5.0

DIFFICULTY

3.0

SOC317

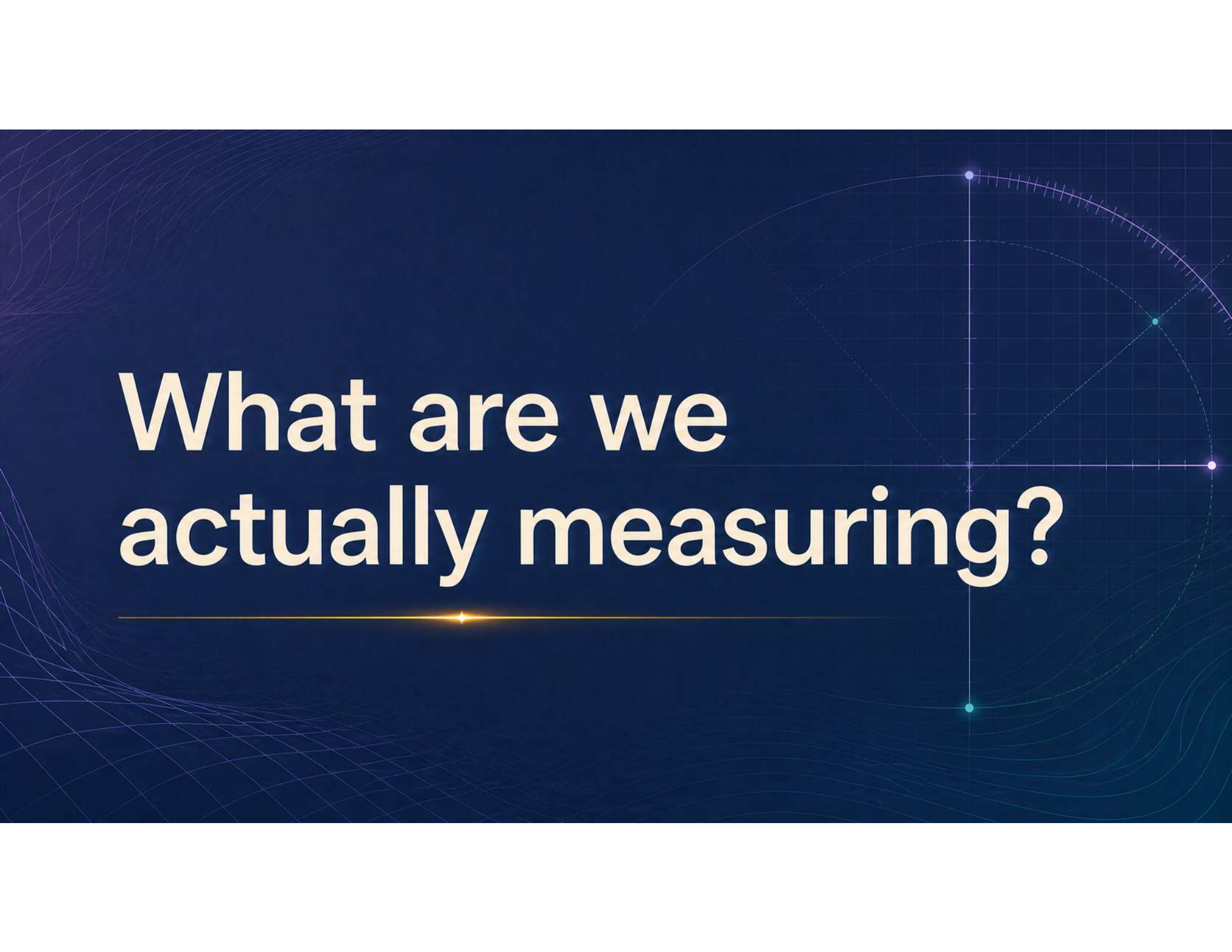
Aug 12th, 2004

For Credit: **Yes** Attendance: **Not Mandatory** Would Take Again: **Yes** Textbook: **Yes**

Big up to the selektah massive inside! Rinse out Prof. [REDACTED] propa...gunshots! boh! boh!
maximum Reeerespect....

Helpful  0  0



The background is a dark blue field with a subtle grid pattern. On the right side, there are several glowing points in light blue and purple, connected by thin, curved lines that suggest a network or a path. A horizontal line with a small, bright orange-yellow glow is positioned below the text.

**What are we
actually measuring?**

“ Scores averaged across an ill-defined assortment of items offer no basis for knowing what is being measured.”

Marsh & Roche — American Psychologist, 1997 · p. 1187

QUALITY

3.0

Oct 30th, 2008

Textbook: No

DIFFICULTY

5.0

dude is huge and i mean huge and is a rude guy but he knows alot about politics

Helpful  1  2



QUALITY

2.5

Aug 27th, 2011

Textbook: Yes

DIFFICULTY

4.0

Oh man. I don't know how she got such good ratings. Her voice makes you want to... hurt... things. every seen mrs. doubtfire? she's like a mix of that and julia child. makes it hard to focus and i think she does it on purpose! i don't mean to be hateful... but you need the facts. don't take her. definitely. I even got a B and i'm sayin this!

Helpful  0  0



QUALITY

5.0

Aug 12th, 2004

For Credit: Yes Attendance: Not Mandatory Would Take Again: Yes Textbook: Yes

DIFFICULTY

3.0

"Big up to the selektah massive inside! Rinse out Prof. [redacted] propa...gunshots! bohl bohl! maximum Reespect...."

Helpful  0  0



Why ratings can be systematically contaminated

Wongsurawat (2011)



Pedagogically
irrelevant comments
introduce
measurement error.

Schiekirka & Raupach (2015)



Student and
process factors
systematically
confound ratings.

Li et al. (2025)



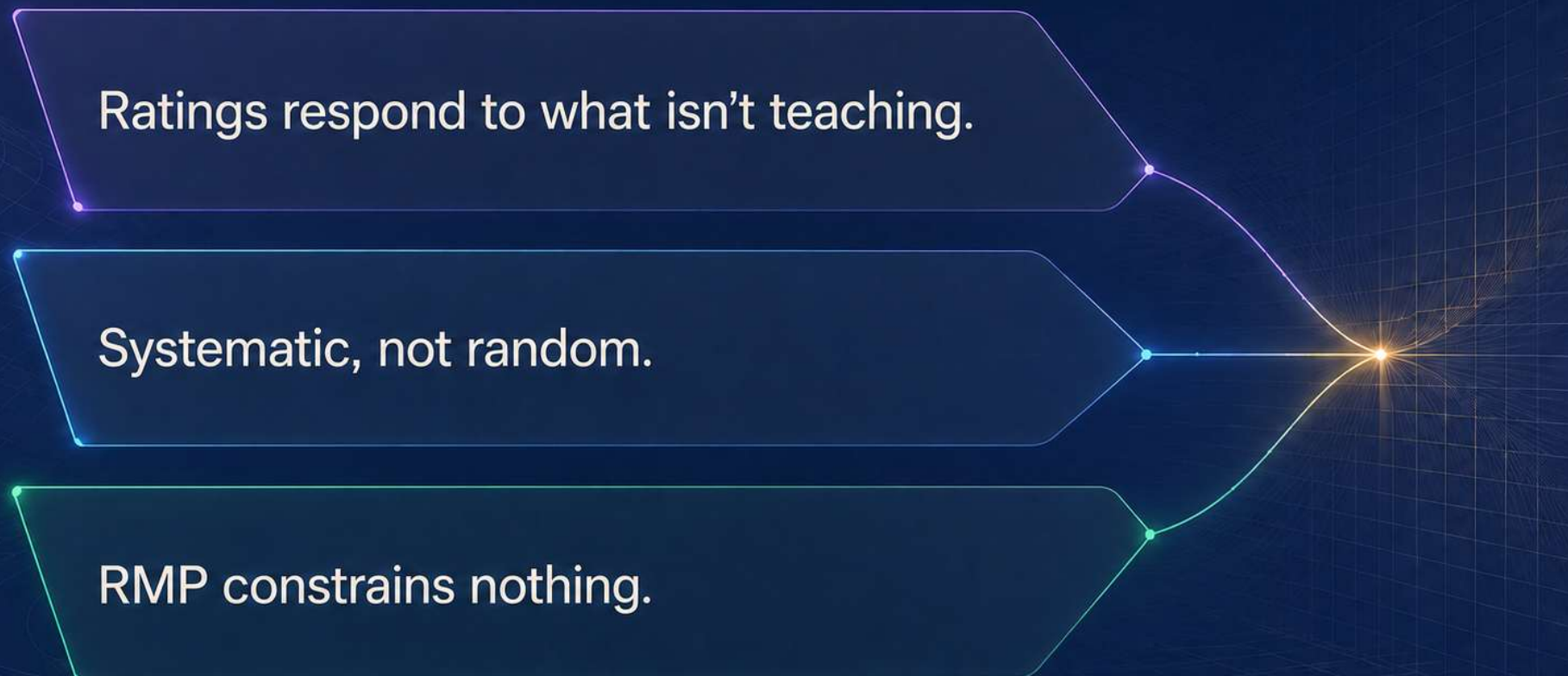
Emotional
interaction can
overshadow
teaching quality.

Three literatures, one finding

Ratings respond to what isn't teaching.

Systematic, not random.

RMP constrains nothing.



Possible Solutions

Constrain the Instrument

Narrow, targeted items.

Removes noise by construction.

Cost: Loses the expressiveness that makes open ended feedback worth collecting; can't be applied to existing data.

Discard bad reviews

Filter “low quality” comments.

Removes noise by exclusion.

Cost: Discards signal along with noise. Who decides what's “bad”?

Down-weight by content

Keep every review, but reduce the influence of irrelevant content.

Attenuates noise, while aiming to preserve signal.

Bounded, auditable, contextual to each review.

Down-weighting is the principled choice



Cronbach (1951) — Reduce error variance, don't remove observations.



Huber (1964) — Down-weighting over deletion for stability.



Marsh & Roche (1997) — Don't throw the validity baby out.

The corpus



17,127 reviews ·
1,029 professors



40.1 words on
average — short, informal

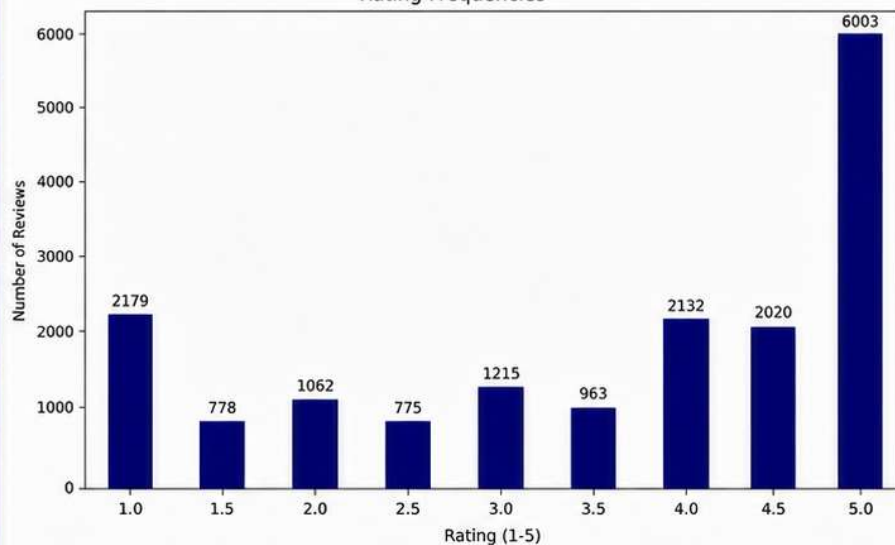


Heavily skewed
toward 5 stars

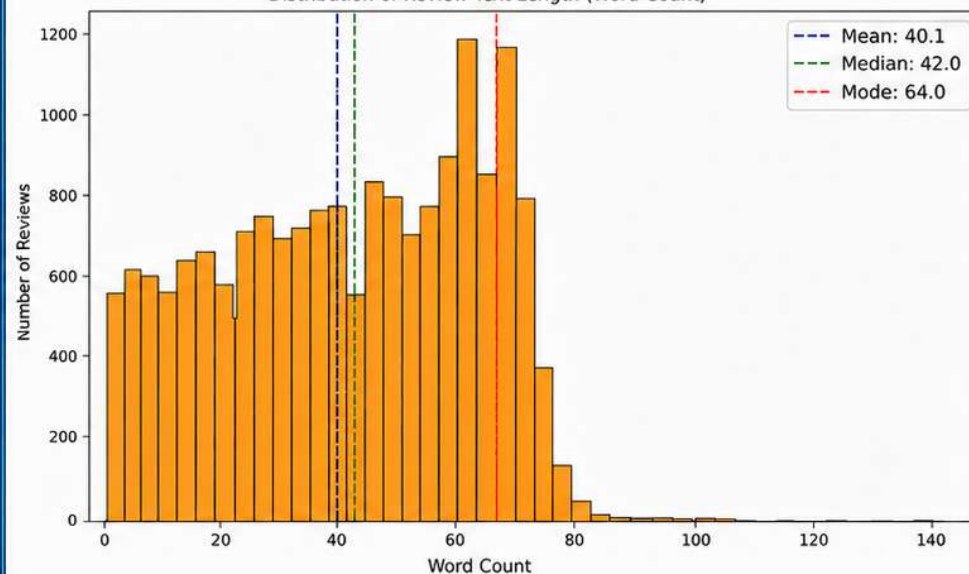


Open dataset (He, 2020) —
no scraping

Rating Frequencies



Distribution of Review Text Length (Word Count)



Research Questions



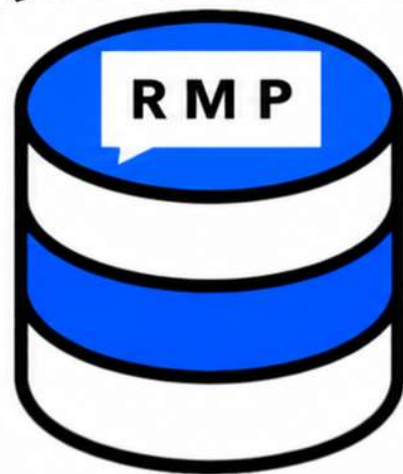
RQ1 — Does pedagogical emotion drive ratings?



RQ2 — Does off-topic emotion drive ratings?



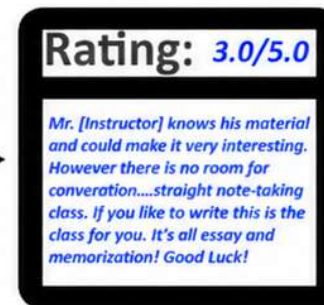
RQ3 — Can we attenuate the noise and keep the signal?



RateMyProfessor
Dataset



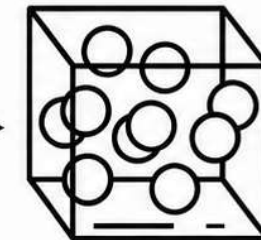
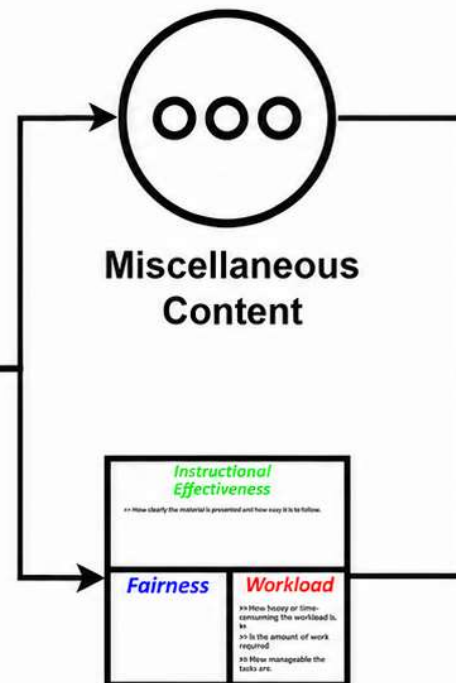
Data Cleaning



Individual
Student
Reviews



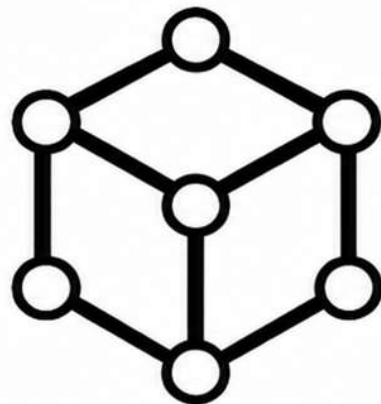
Aspect Term
Categorization



Pedagogical
Content Density
Metric (0.0 - 1.0)



HUGGING FACE



**Multi-Label
RoBERTa GoEmotions Model**

***Instructional
Effectiveness***



Fairness

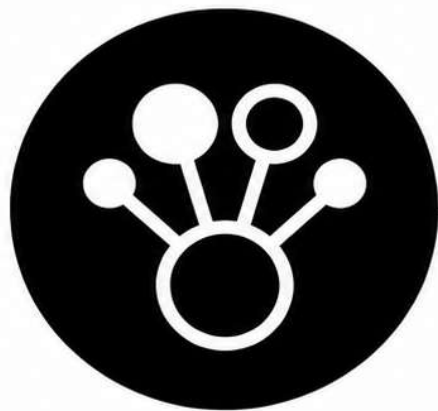


Workload



Miscellaneous

**Extracting Emotions from Student Reviews
For Each Topic**



**CatBoost
Regression Model**



<i>Instructional Effectiveness</i>	😊 😞 😞 😞 😞
<i>Fairness</i>	😞 😞 😞 😞 😞
<i>Workload</i>	😞 😞 😞 😞 😞
<i>Miscellaneous</i>	😞 😞 😞 😞 😞



**Numerical
Rating**
2.5 ★★★★★

Train CatBoost Model to Predict Ratings

Baseline Emotions

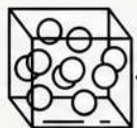
Instructional Effectiveness (green text) 5 smiley faces

Fairness (blue text) 5 neutral faces

Workload (red text) 5 sad faces

Miscellaneous (black text) 5 sad faces

Pedagogical Statement Density Metric (0.0 - 1.0)



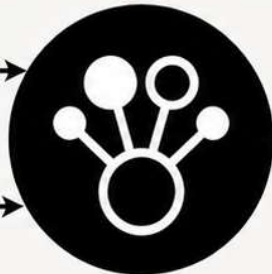
Instructional Effectiveness (green text) 5 smiley faces

Fairness (blue text) 5 neutral faces

Workload (red text) 5 sad faces

Misc. (black text) 5 sad faces

Down-Weighted Misc. Emotions (Attenuated)



Trained CatBoost Model

Raw Student Ratings

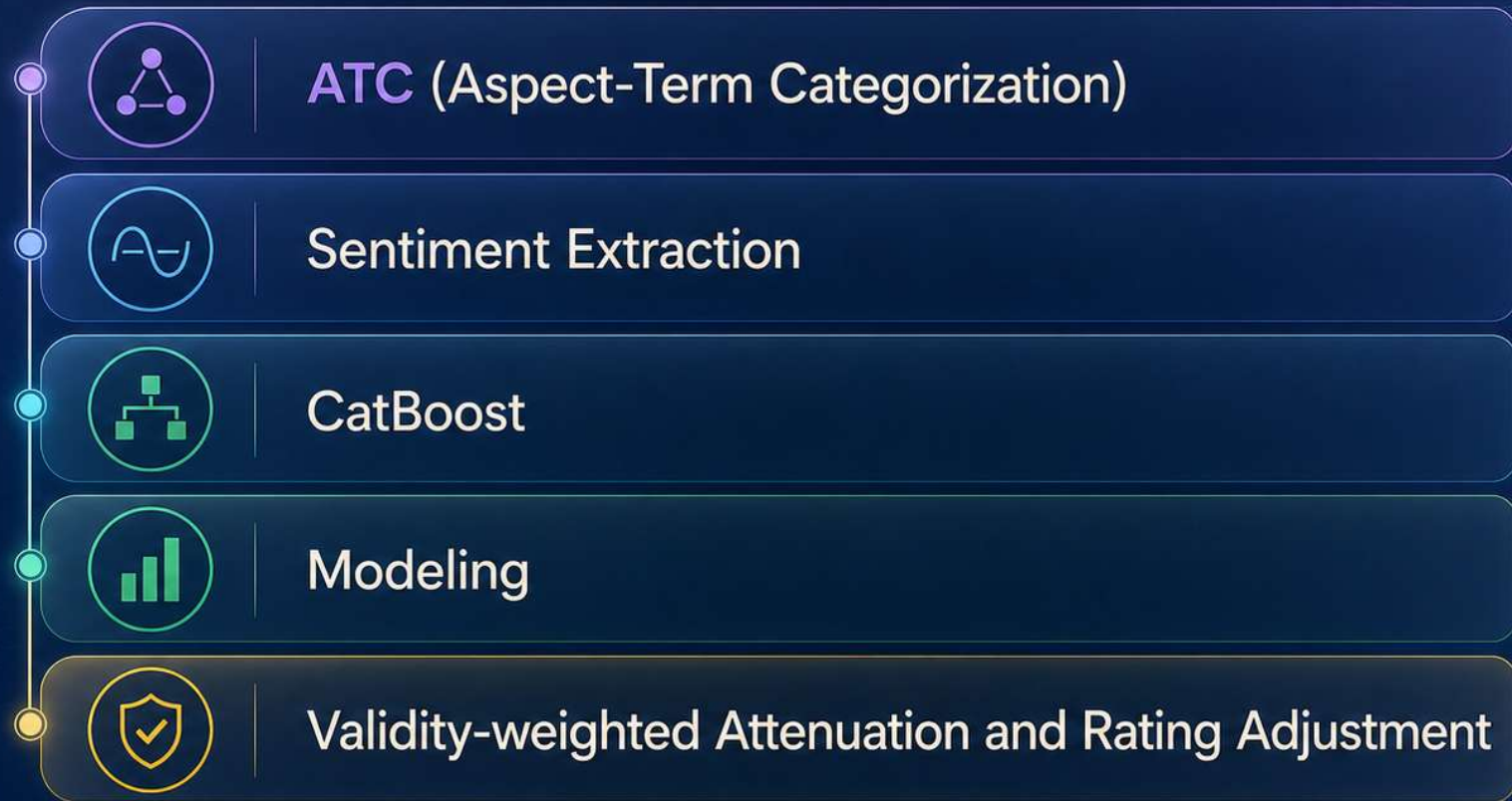


$$\left(\hat{Y}^{\text{attenuated}} - \hat{Y}^{\text{baseline}} \right) + \hat{Y}^{\text{raw}}$$



Adjusted Ratings

In Greater Detail



ATC



Miscellaneous

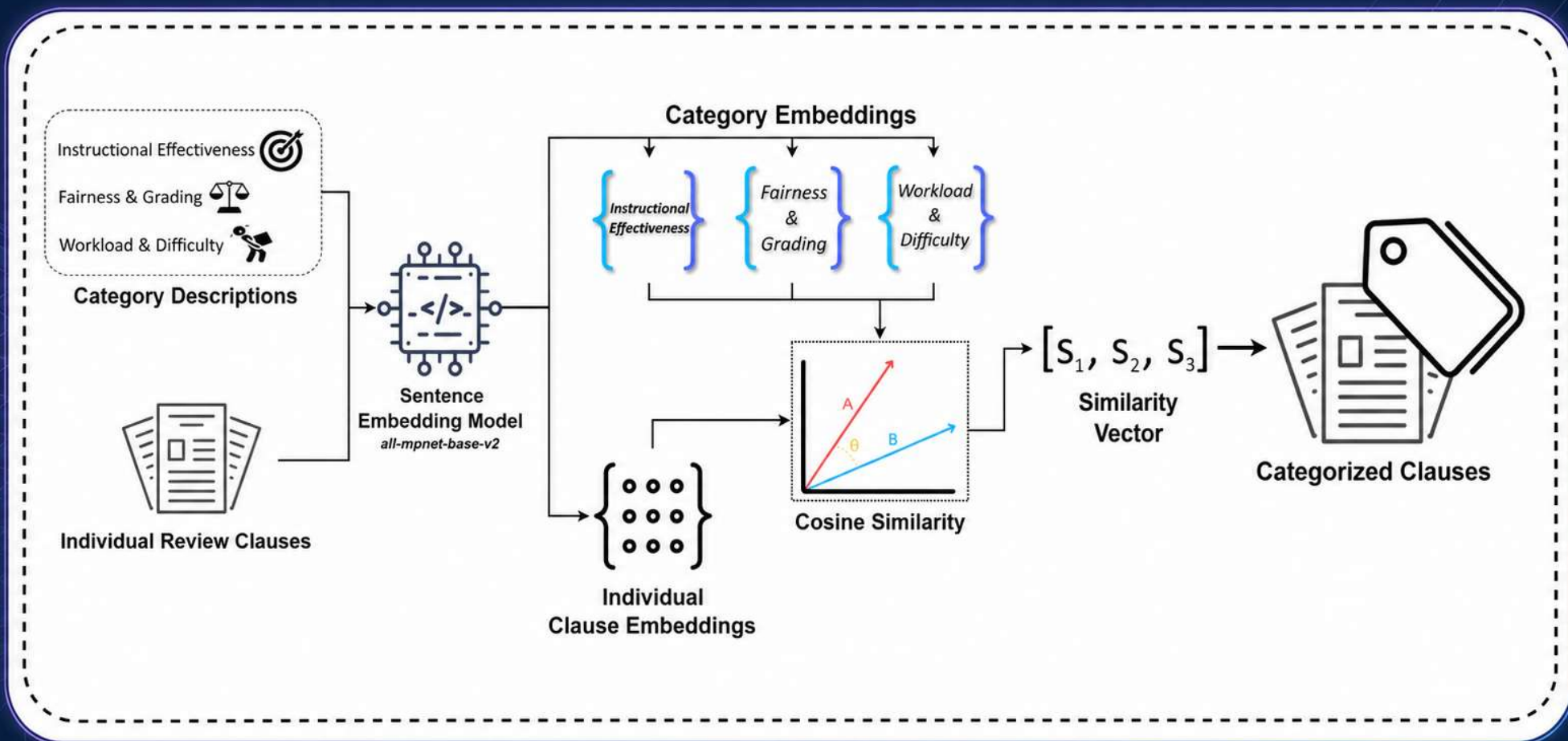


Density

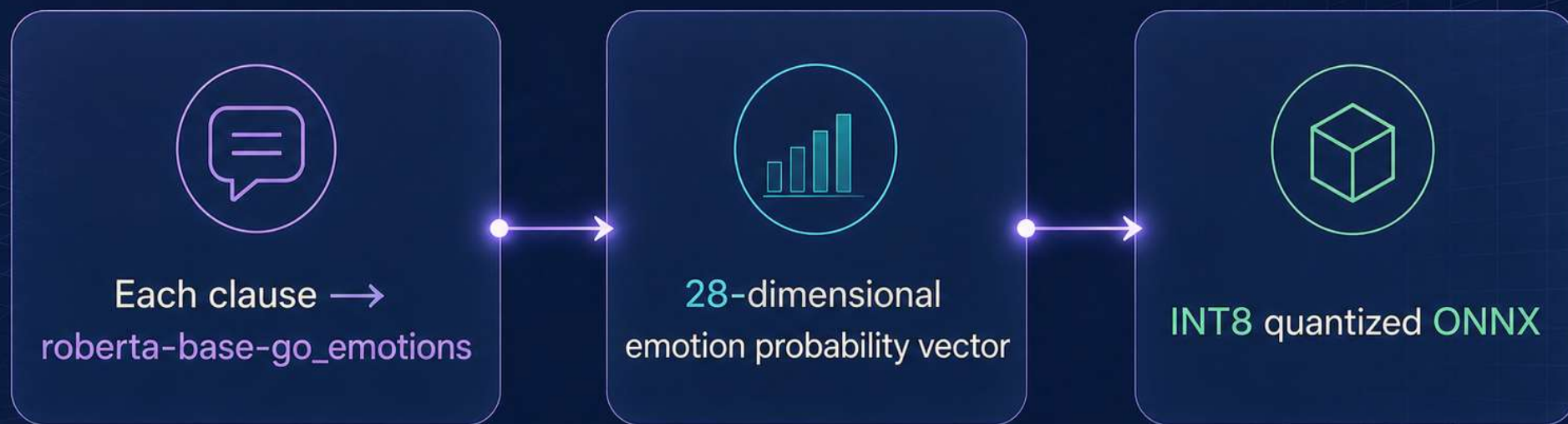
The share of a review aimed at Miscellaneous

1.0 → fully off-topic

0 → fully on-topic



Sentiment Extraction



Feature Aggregation per Review



One vector per review, across all categories



Average emotions within each category



Impute 0 where a category is absent

Feature Aggregation Cont.

Final feature vector per review =
 $27 \times 4 + (D_{misc}, D_{eff}, D_{work}, D_{fair}) = 112$ features

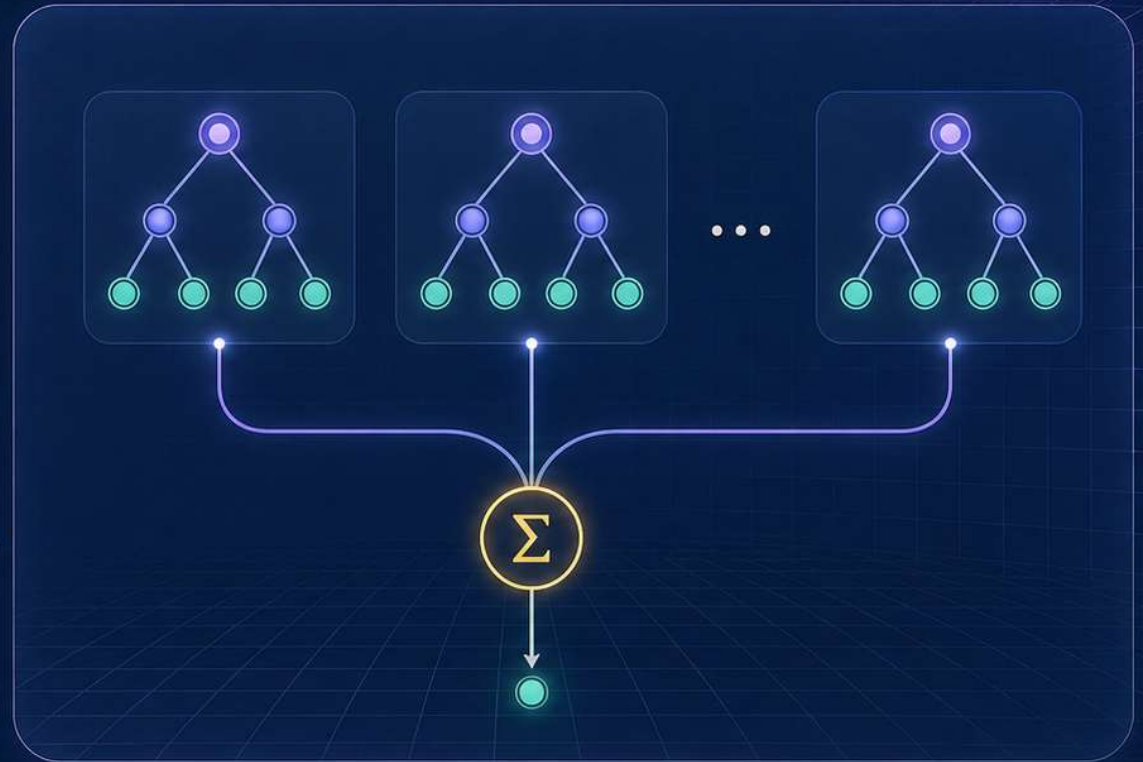
D_{misc} | = Miscellaneous density

D_{eff} | = Instructional Effectiveness density

etc.

CatBoost

- Modeling
- Gradient-boosted trees
- Strong on sparse, non-linear data
- Professor-level grouped split · 5-fold CV
- Model and baseline predictions saved

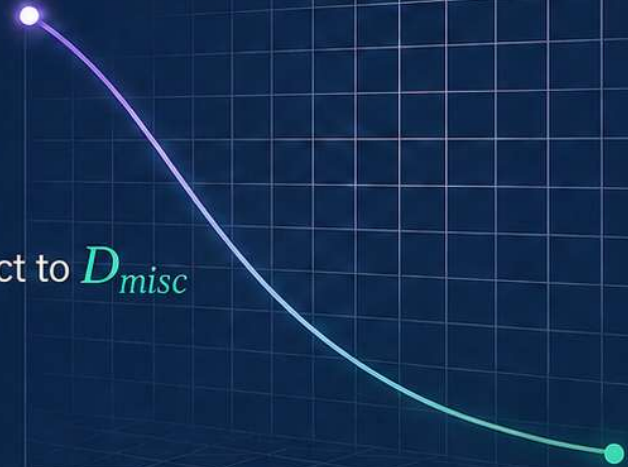


Validity-Weighted Attenuation

Misc emotions are down-weighted by ζ

$$\zeta = 1 - (D_{misc})^s$$

- ⋄  S controls the non-linearity of ζ
- ⋄  As D_{misc} increases, ζ decreases
- ⋄  *Misc* emotions are proportionally down-weighted with respect to D_{misc}
- ⋄  All other features are preserved.



Validity-Weighted Attenuation Cont.

$$\hat{y}^{\text{adjusted}} = y^{\text{raw}} + (\hat{y}^{\text{attenuated}} - \hat{y}^{\text{baseline}})$$

- 1  Attenuated vector back into CatBoost → new prediction
- 2  $\Delta = \text{attenuated prediction} - \text{baseline prediction}$
- 3  Add Δ to the student's original rating
- 4  Ta Da!

Worked Example — Review #8113

Section 1: Clause Extraction and Categorization

Clause	Topic	Sim	Top 2 Emotions
dude is huge	Miscellaneous	0.030	admiration=0.044, approval=0.024
i mean huge	Miscellaneous	-0.010	approval=0.018, annoyance=0.015
is a rude guy	Miscellaneous	0.200	anger=0.567, annoyance=0.361
he knows alot about politics	Instr. Effectiveness	0.270	approval=0.293, admiration=0.031

Section 2: Baseline vs. Attenuated Feature Vector

Topic	Baseline Top 4 Emotions	Down-Weighted Top 4 Emotions
Instr. Effectiveness	approval=0.293, admiration=0.031, realization=0.016, optimism=0.004	approval=0.293, admiration=0.031, realization=0.016, optimism=0.004
Fairness	—	—
Workload	—	—
Miscellaneous	anger=0.192, annoyance=0.129, admiration=0.016, approval=0.016	anger=0.056, annoyance=0.038, admiration=0.005, approval=0.005

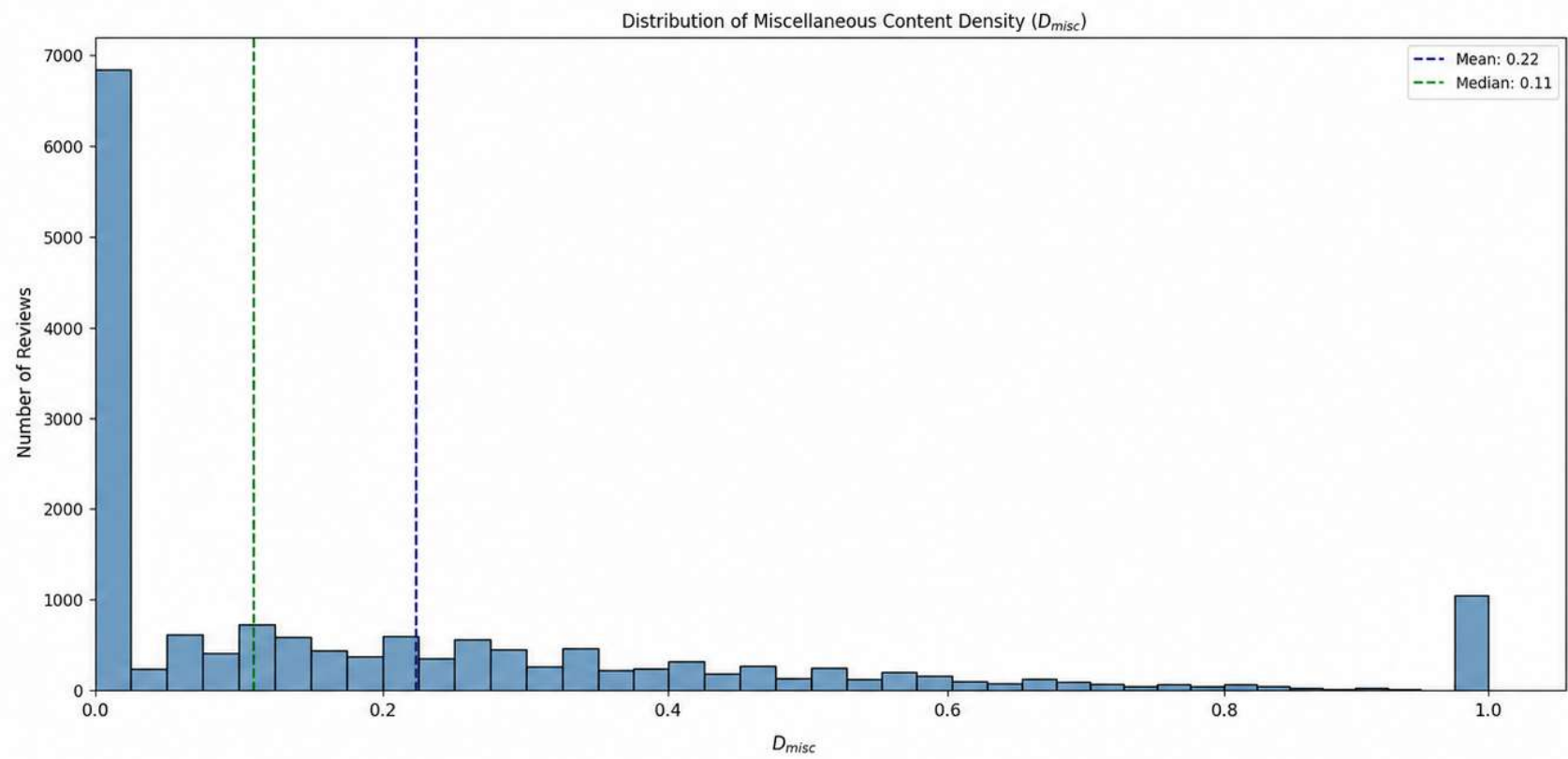
Section 3: Validity-Weighted Attenuation

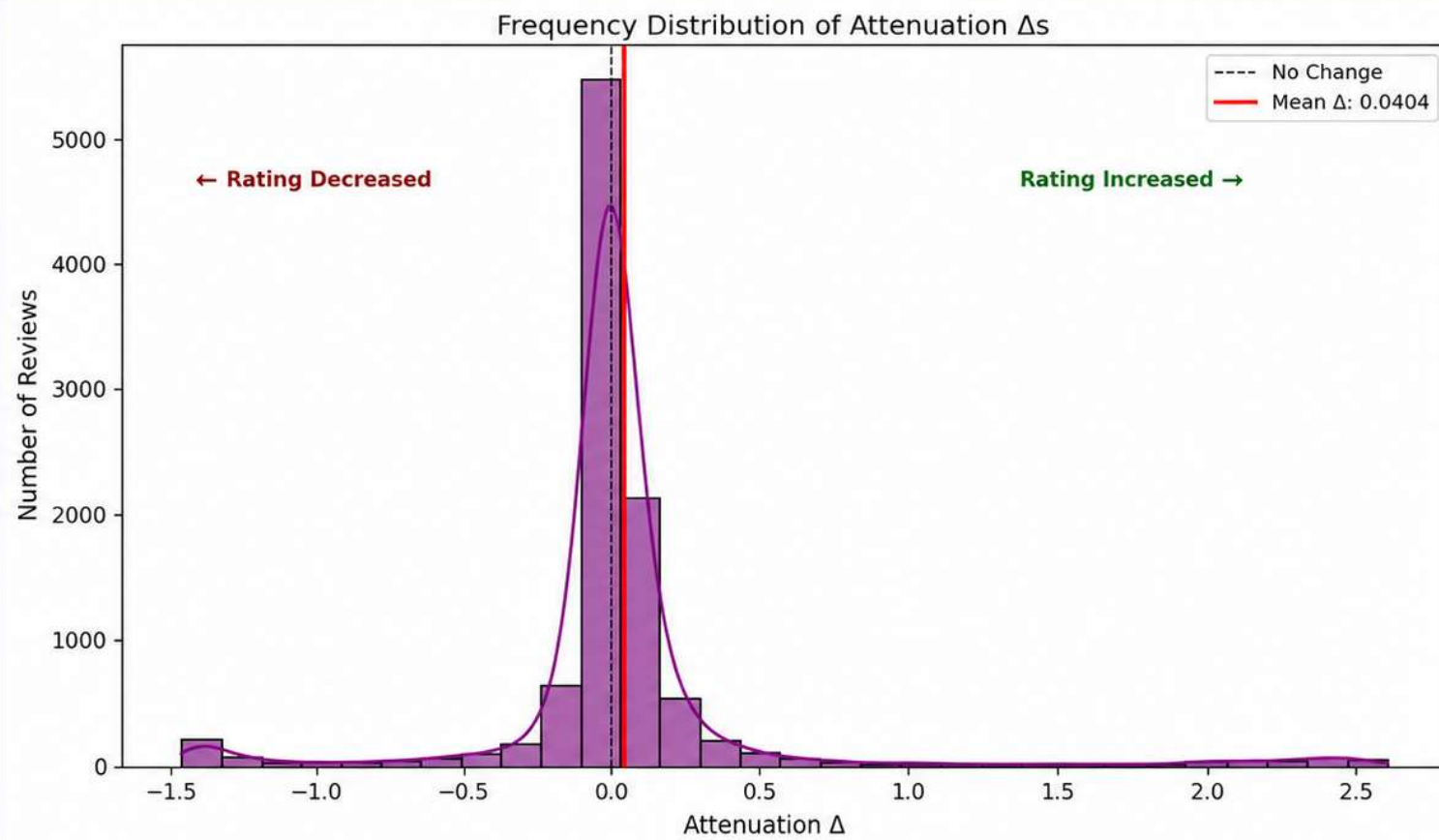
D_{misc}	ζ	Raw	Baseline	Attenuated	Δ	Adjusted
0.670	0.291	3.0	2.506	3.159	+0.653	3.653

Three of four clauses are Miscellaneous ($D_{\text{misc}}=0.670$), yielding $\zeta=0.291$. Miscellaneous emotions are down-weighted by approximately 71%. The instructional effectiveness signal remains unchanged. The adjusted rating increases by 0.653.

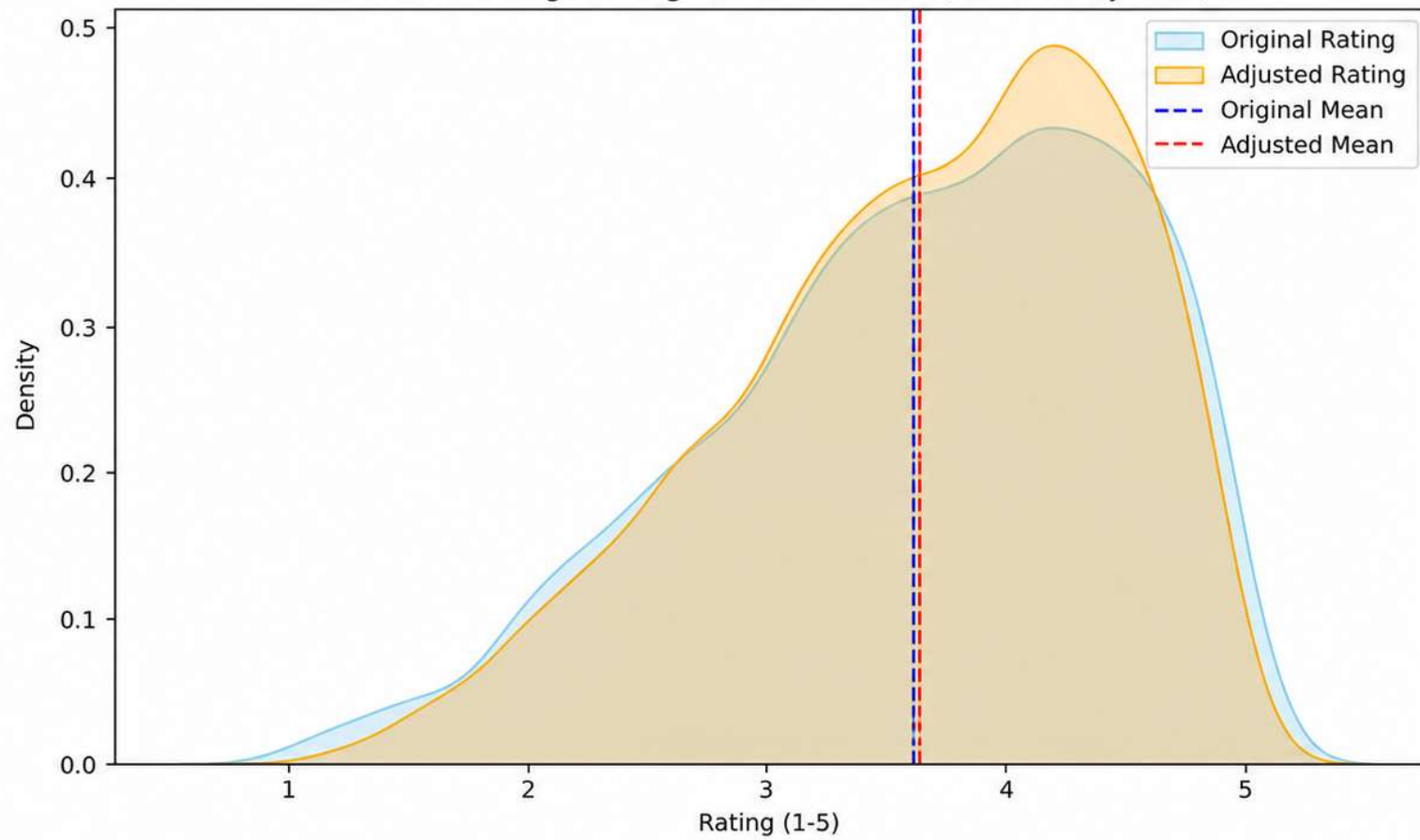
Results

The background is a dark blue gradient with a subtle grid pattern. A glowing purple arc curves from the bottom left towards the top right, ending in a small, bright purple dot. A horizontal line with a bright yellow starburst effect is positioned below the word "Results".

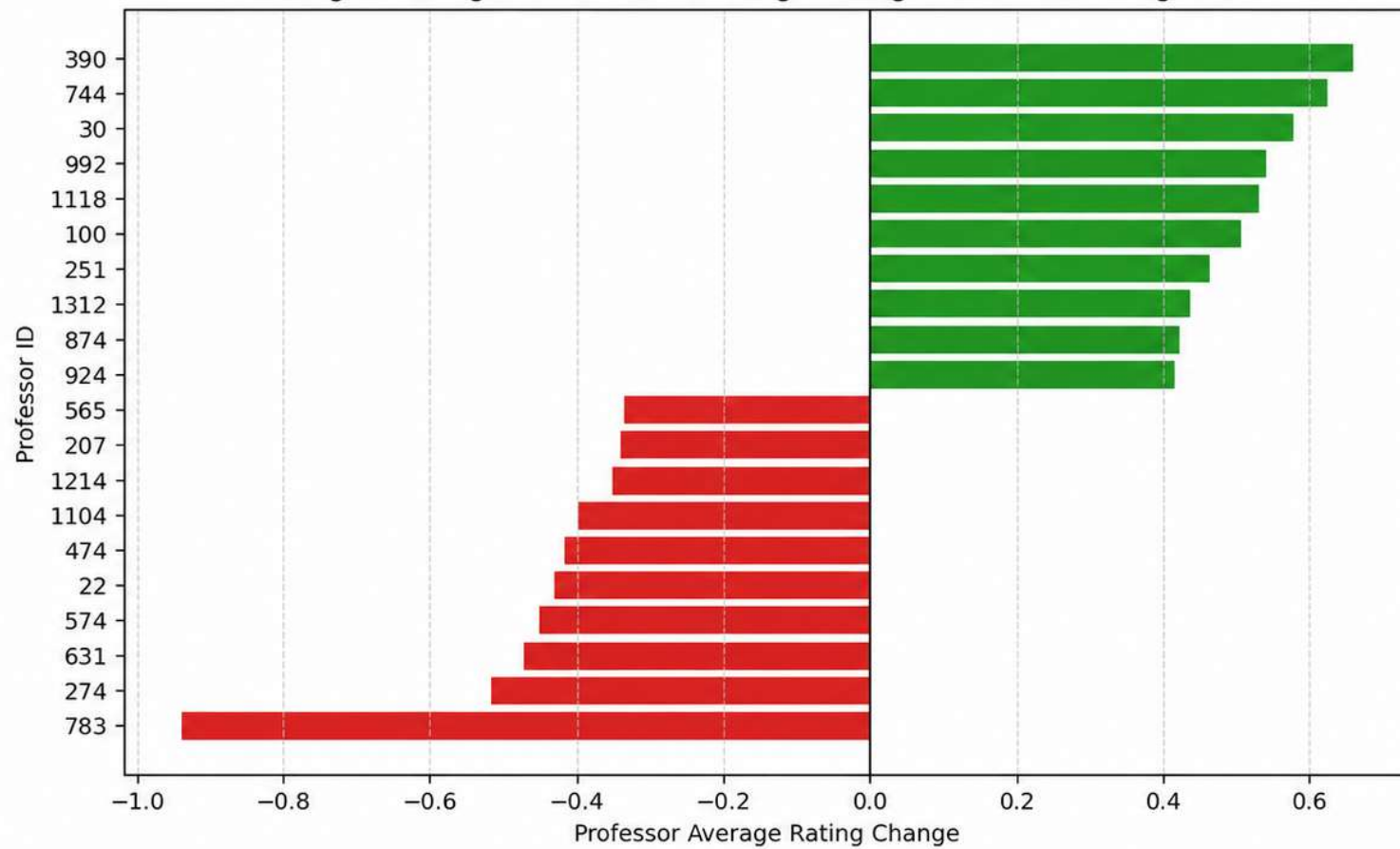




Professor Average Rating Distribution Shift (Raw vs. Adjusted)



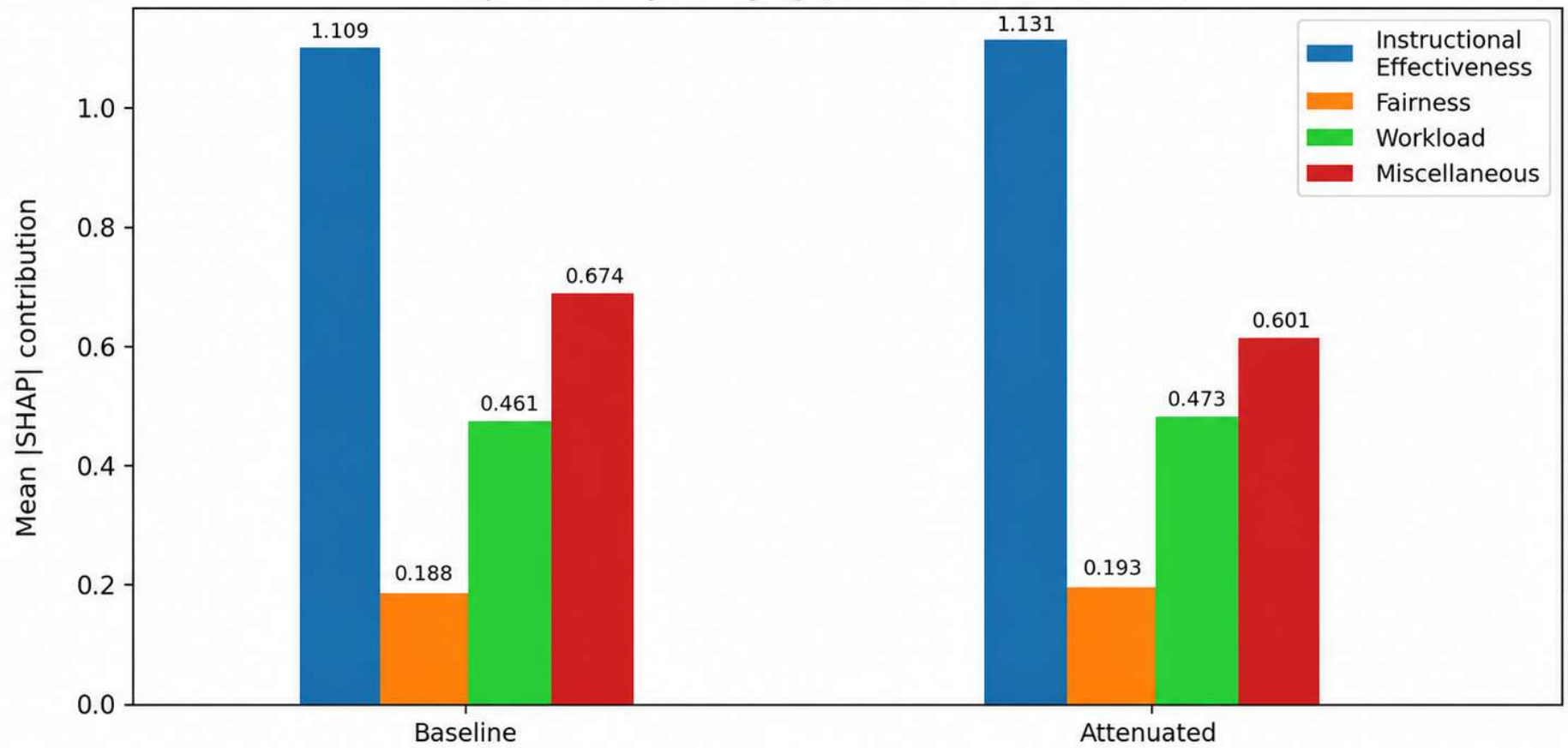
Largest Changes in Professor Average Ratings (Positive and Negative)



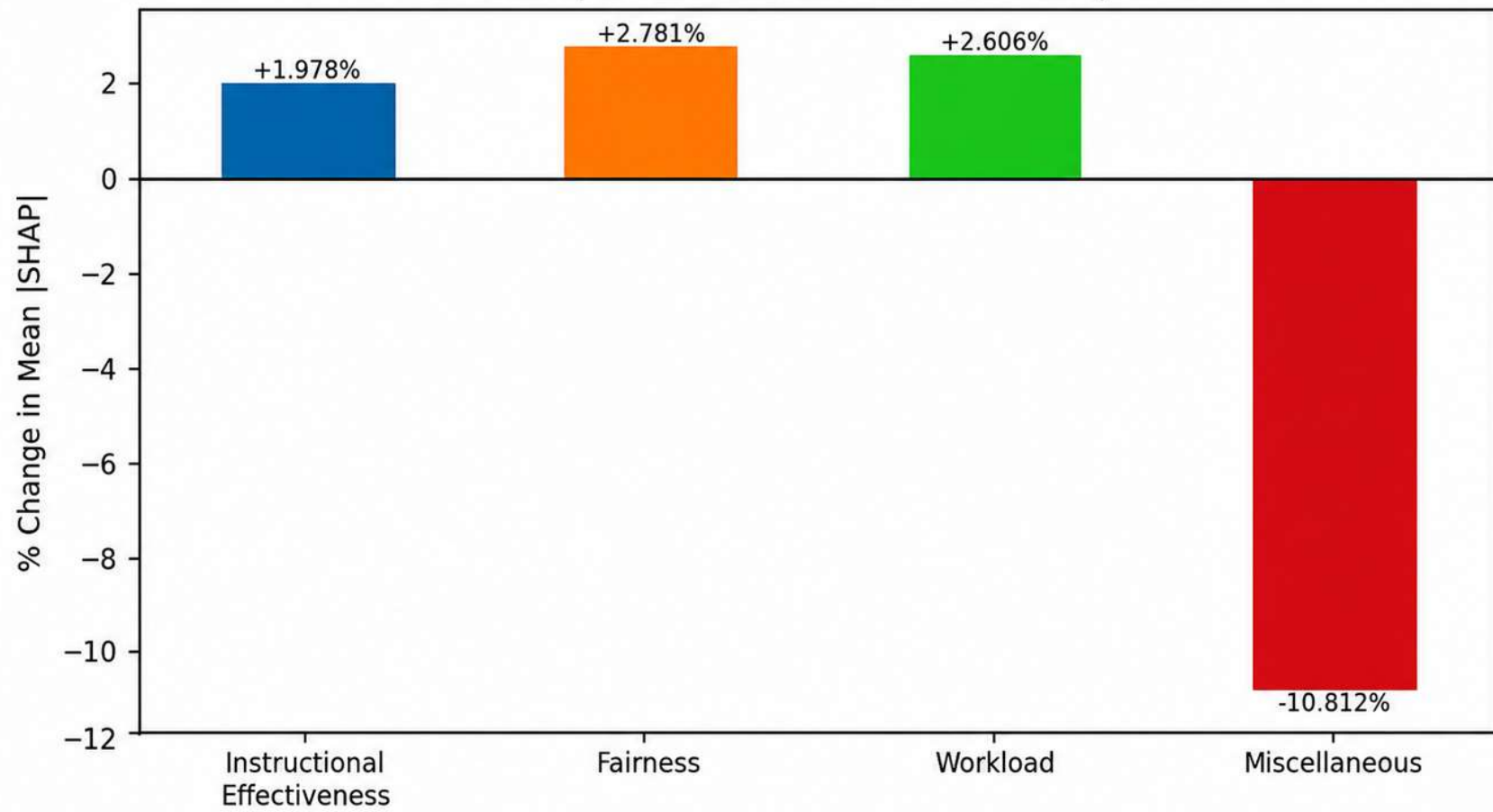
Validation



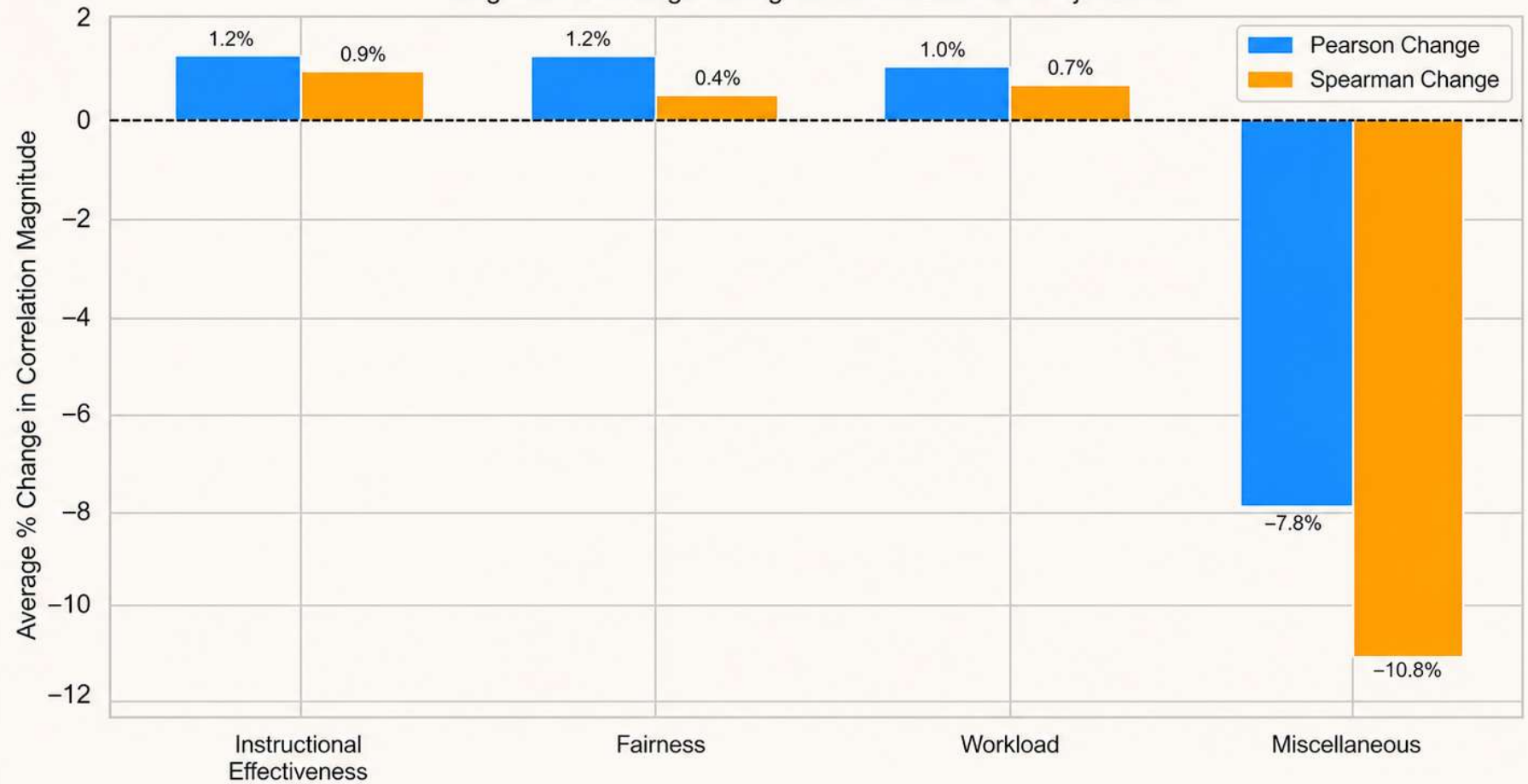
Importance by Category (Baseline vs Attenuated)



Relative Impact of Attenuation on Feature Importance



Weighted Percentage Change in Correlations After Adjustment



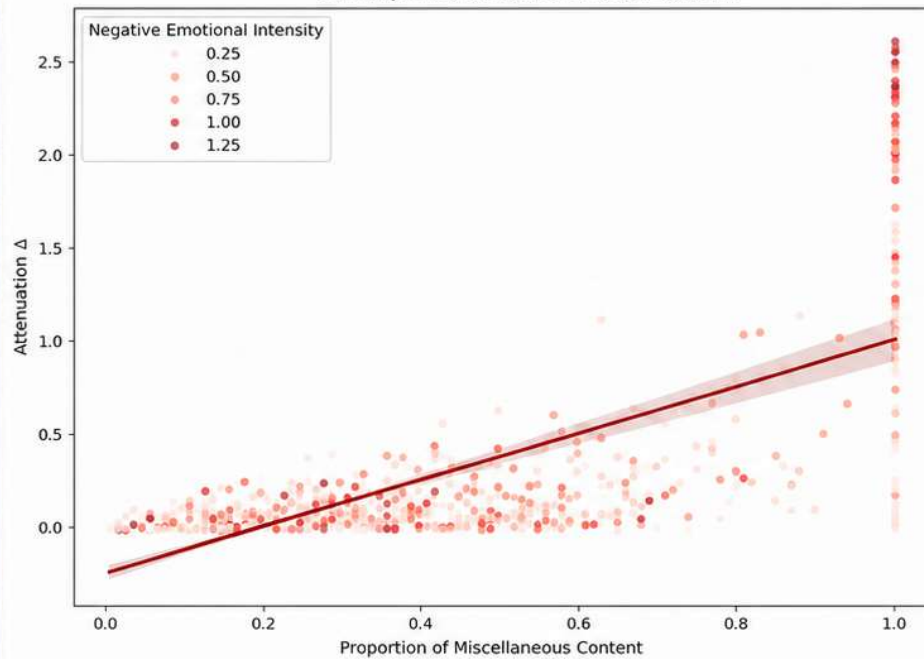
$\Delta+$ Versus

	Pearson r	Spearman ρ
D_{misc}	0.6956	0.6218
$Neg. E_{misc}$	0.3600	0.4185

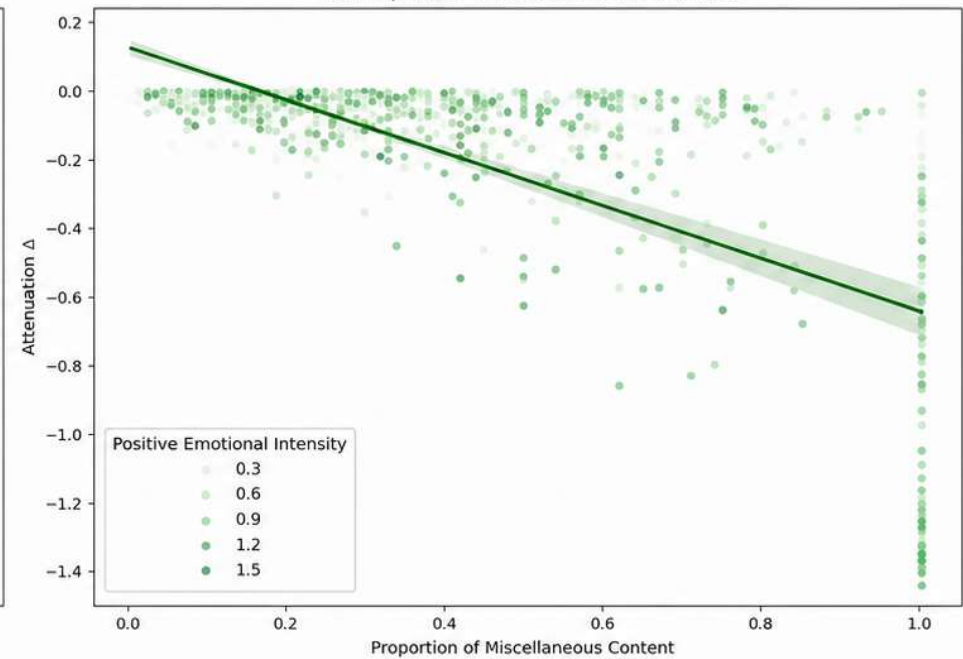
$\Delta-$ Versus

	Pearson r	Spearman ρ
D_{misc}	-0.6966	-0.5607
$Neg. E_{misc}$	-0.2258	-0.1063

Positive Attenuation Δ 's (Rating Increased)
vs. Proportion of Miscellaneous Content



Negative Attenuation Δ 's (Rating Decreased)
vs. Proportion of Miscellaneous Content



External Expert Validation

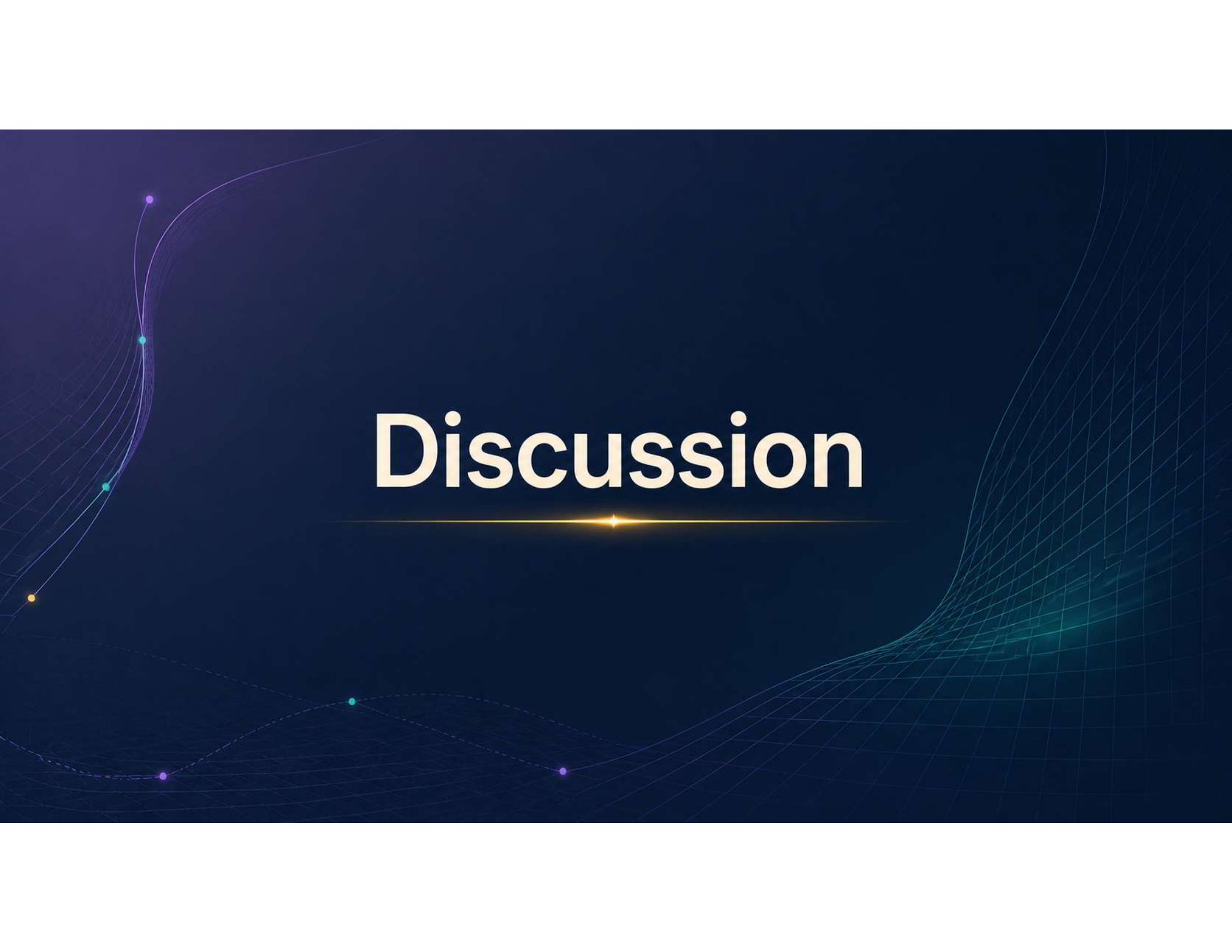


Pairwise comparison

Do our relative adjustments match an expert's?

Condition	Number of pairs	Correct	Unsure	Accuracy	Significance
$ \Delta_1 - \Delta_2 \geq 0.5$	45	40	3	0.889	$p < 0.001$
$ \Delta_1 - \Delta_2 \leq 0.4$	29	22	1	0.759	$p < 0.005$

Discussion

The background is a dark blue gradient with abstract, glowing lines and dots in shades of purple, teal, and yellow. A horizontal line with a bright yellow glow is positioned below the word "Discussion".

RQ1



Pedagogical emotion drives ratings.

**Instructional Effectiveness
dominates.**

RQ2

Off-topic emotion drives ratings too.

Misc ranks 2nd as a category — 3rd by individual feature.

And it **contaminates** the whole scale.

RQ3

- Importance shifts off *Misc*, onto pedagogy.
- Correlations separate cleanly.

Mean $|\Delta| = 0.207$ — a scalpel, not a hammer.

RQ3 Cont.



Mechanism implemented in alignment with theory (Messick, Huber, Cronbach).



Interpretability as a constraint.



Practical Implications



Trustworthier
scores on
platforms like RMP



A direction for
formal SET systems



Insight for
faculty development



Portable to other
feedback domains

Limitations

We don't know if the method works,
if the data is clean, or if the math is right,
but I can tell you with absolute, 100%
certainty exactly how we ruined it!

Limitations



No ground truth for teaching quality



One platform, one corpus



Preliminary expert validation



Expert test measures direction, not magnitude



Errors Cascade

Limitations Cont.



ATC is broad and single-label.



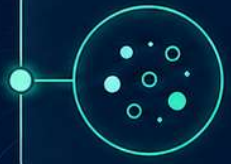
RoBERTa-GoEmotions was trained on Reddit.



Collapse point at $D_{misc} = 1$.



Neutral-but-off-topic is invisible.



Just one flavour of noise (off-topic).



Efficiency-first choices cap the ceiling.

Conclusion

The background is a dark blue gradient. On the left, there is a faint grid pattern. On the right, there are several glowing, curved lines in shades of purple and blue, some of which are composed of small dots. A bright, horizontal light flare is positioned directly beneath the word 'Conclusion'.



Contributions



Intervene, don't just detect



Psychometric theory made computational



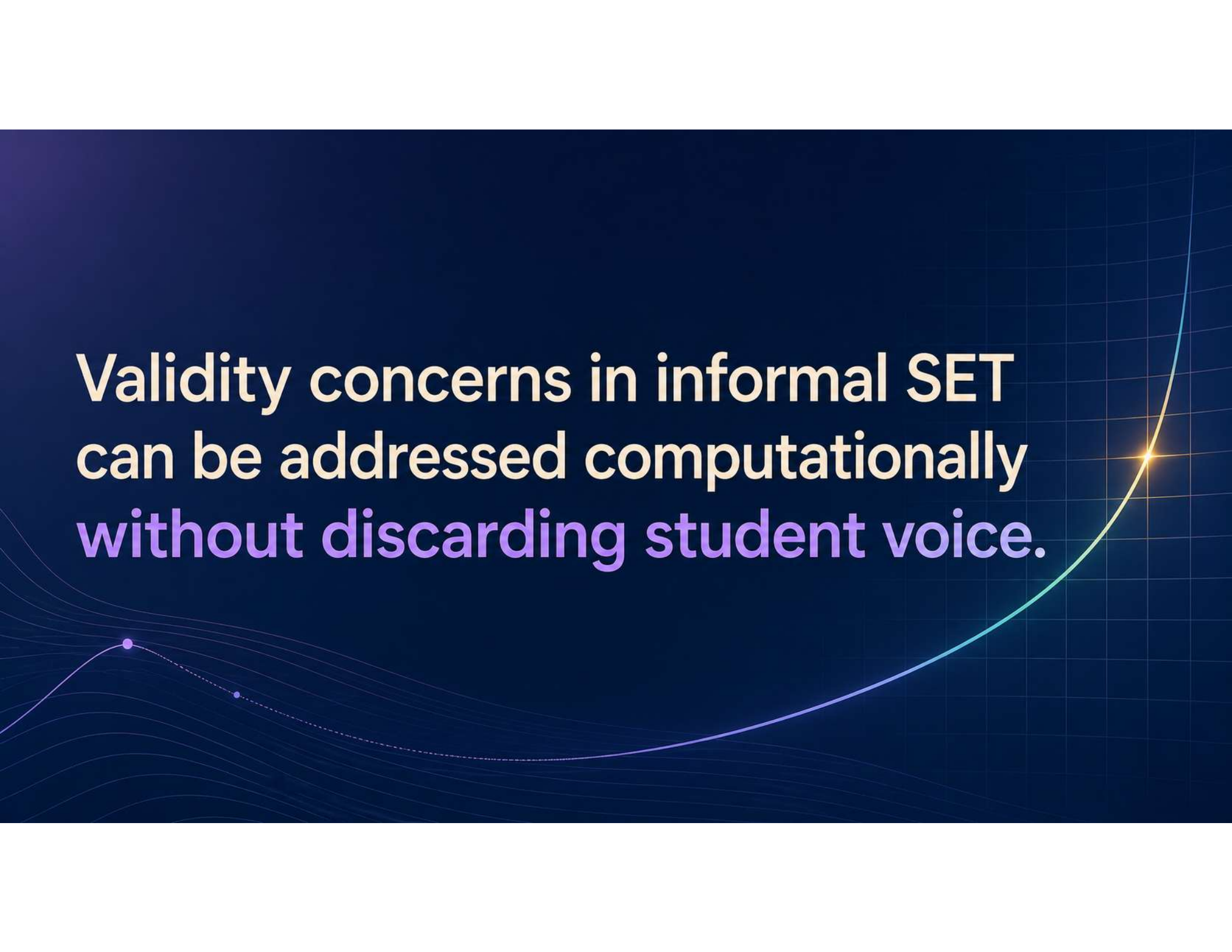
Every adjustment is auditable

Proof-of-concept noise reduction, not a validated measure of teaching quality.

| Possibly extendable to formal SET systems.

What's Next?





**Validity concerns in informal SET
can be addressed computationally
without discarding student voice.**

Acknowledgements

-  AMMCS Participants and Staff
-  Professor Somayeh Fatahi
-  My Father
-  Senior Faculty Experts
-  Laurier REB



Appendix

The background is a dark blue gradient. It features several glowing, wavy lines in shades of purple, teal, and yellow. These lines are composed of many thin, parallel strokes that create a sense of motion and depth. Small, bright dots of the same colors are scattered along these lines, acting as focal points or data points. A horizontal line of light, resembling a lens flare or a starburst, is positioned directly beneath the word 'Appendix'.

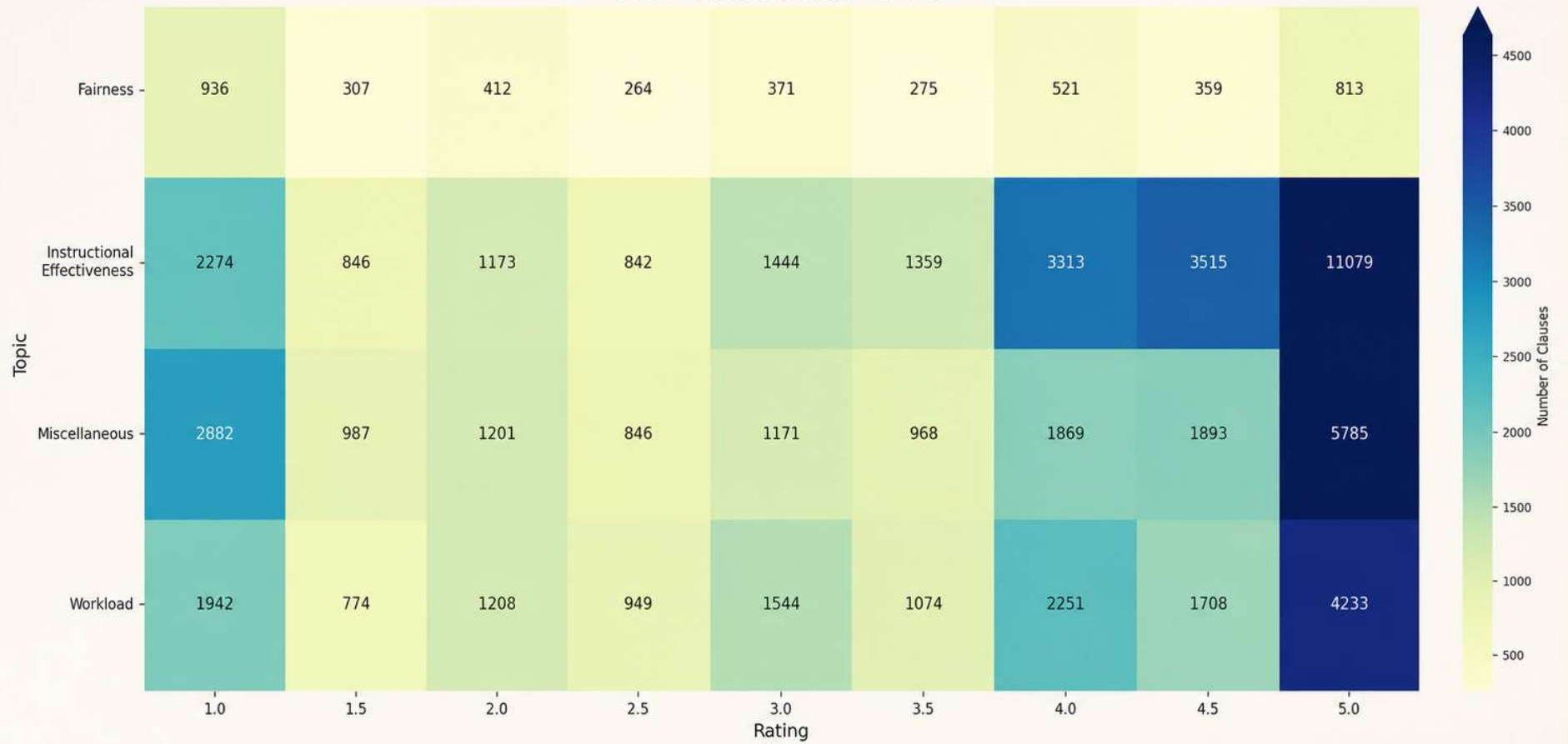
Other non-CatBoost Options

Model	MAE ↓	Spearman ρ ↑	R^2 ↑
Ordinal Regression	0.9509	0.5414	0.0794
Linear Regression	0.7940	0.6875	0.5266
Random Forest	0.6938	0.7289	0.6217
XGBoost	0.6710	0.7329	0.5638

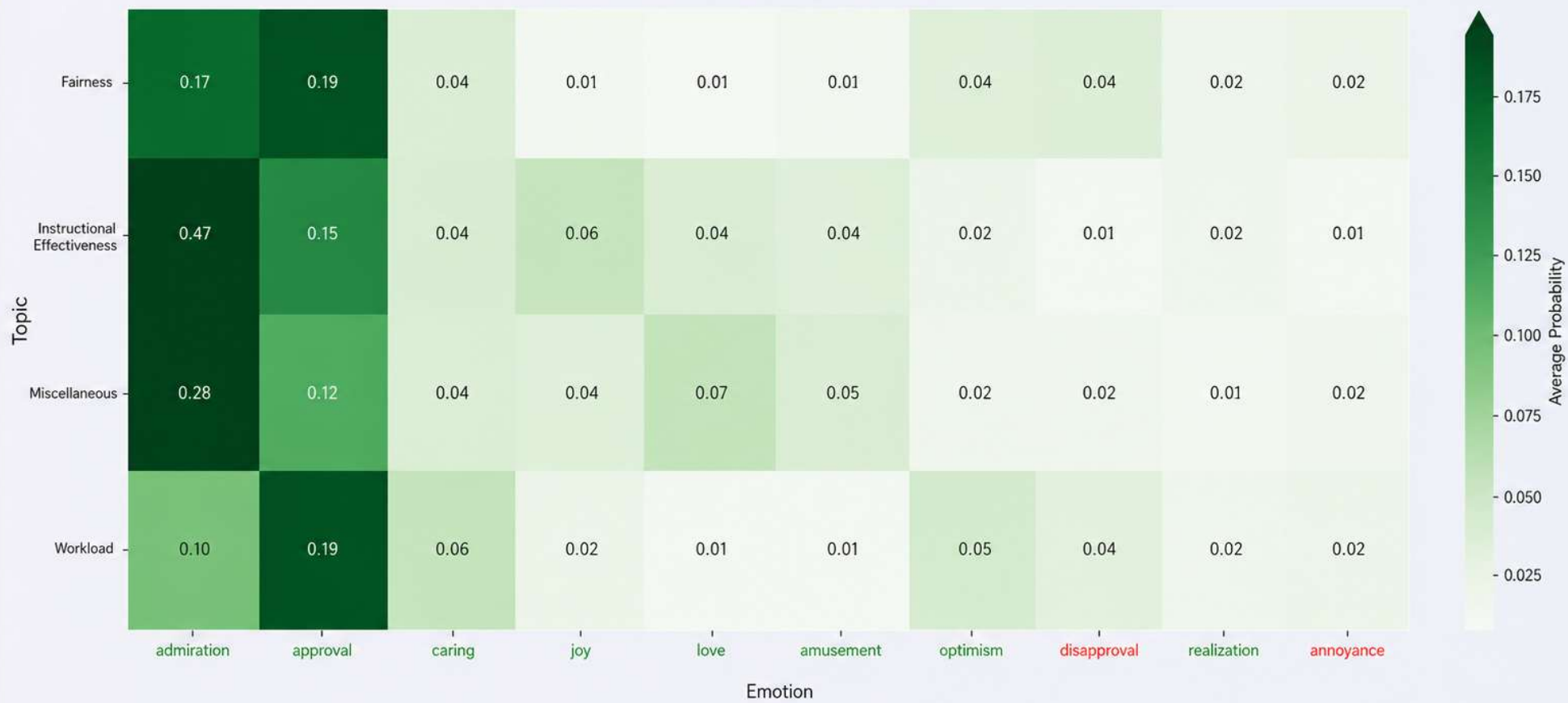
Final Choice	MAE ↓	Spearman ρ ↑	R^2 ↑
CatBoost	0.6369	0.7355	0.6140

Fold	MAE	Spearman ρ ↑	R^2 ↑
1	0.6691	0.7102	0.5837
2	0.6195	0.7424	0.6431
3	0.6052	0.7462	0.6476
4	0.6328	0.7282	0.6262
5	0.6236	0.7241	0.6282
Mean	0.63	0.7302	0.6258

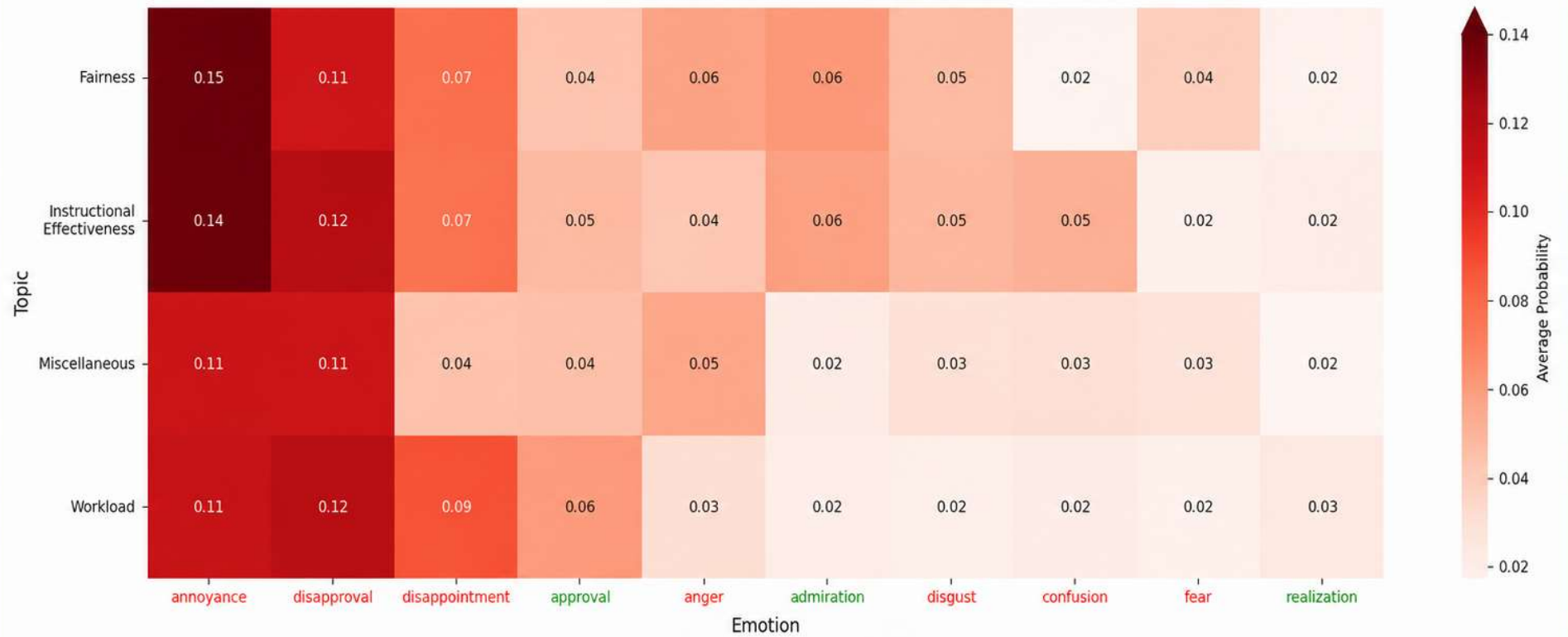
Clause Topics vs Rating: Heatmap



5-Star Clauses: Emotion Heatmap



1-Star Clauses: Emotion Heatmap



Positive
Instructional
Effectiveness

Negative
Instructional
Effectiveness

Negative
Miscellaneous

Negative
Workload

Positive
Miscellaneous

Negative
Fairness

Positive
Workload

Positive
Fairness

